

Literature-Implied Status Note for arXiv:1406.3842

run fa_banach.001, agent_lane.05

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Status. `literature_implied_answer` (preprint claim; needs human verification).

Source/open-problem metadata. Michael Cwikel, Mario Milman, and Richard Rochberg, “A brief survey of Nigel Kalton’s work on interpolation and related topics,” arXiv:1406.3842, posted 2014-06-15. On page 2, Section 2, the paper asks whether one-sided compactness propagates under interpolation: if an operator is bounded on both endpoints A_i , $i = 0, 1$, and compact on A_0 , must it be compact on the interpolation space A_* ? The paper says that for the complex method the question “remains unanswered.”

Supporting-answer metadata. Evgeniy Pustylnik, “A new approach to interpolation of compact linear operators,” arXiv:2605.00530, posted 2026-05-01. On page 2, Pustylnik formulates one-sided compactness for embeddings $A_i \hookrightarrow B_i$, assumes the interpolation estimate

$$\|T\|_{A \rightarrow B} \leq W(\|T\|_{A_0 \rightarrow B_0}, \|T\|_{A_1 \rightarrow B_1}),$$

where $W(\alpha, \beta) \rightarrow 0$ as $\beta \rightarrow 0$ uniformly for bounded α , and explicitly notes that for the complex interpolation functor $\mathcal{F}_{[\theta]}$, $0 < \theta < 1$, one may take $W(\alpha, \beta) = \alpha^{1-\theta}\beta^\theta$. The Main Theorem on page 2 then asserts compactness of the interpolated embedding. On pages 7–8 the paper passes from embeddings to arbitrary operators by factoring a continuous operator $T : A_0 \rightarrow B_0$, compact $T : A_1 \rightarrow B_1$, through its image spaces.

Identification. The 2014 question is the classical one-sided compactness problem for the complex method. Pustylnik’s theorem is stated with compactness at the second endpoint A_1 ; the 2014 paragraph phrases compactness at A_0 . For the complex method these are equivalent formulations after swapping the couple (A_0, A_1) and replacing θ by $1 - \theta$. Thus, if the proof of arXiv:2605.00530 is accepted, it gives a positive answer to the target question for the complex method.

Provenance limitation. The supporting paper does not explicitly cite arXiv:1406.3842 in the checked source text. This is therefore an agent-identified implication rather than a paper that explicitly announces it is answering the 2014 survey question. The supporting source is also a very recent arXiv preprint that claims a broad resolution of a long-standing problem, so the packet should be human-reviewed before the run treats the problem as settled.

Search evidence. The run indexes were searched for 1406.3842, Nigel Kalton, propagation of compactness, complex method, one-sided compactness, and Pustylnik/arXiv:2605.00530. No prior packet for this exact target was found. External web lookup found the arXiv records for arXiv:1406.3842, arXiv:2109.05968, arXiv:2605.00530, and the related 2014 survey arXiv:1410.4527. The decisive supporting source used here is arXiv:2605.00530.

References

- [1] M. Cwikel, M. Milman, and R. Rochberg, “A brief survey of Nigel Kalton’s work on interpolation and related topics,” arXiv:1406.3842, 2014.
- [2] E. Pustylnik, “A new approach to interpolation of compact linear operators,” arXiv:2605.00530, 2026.